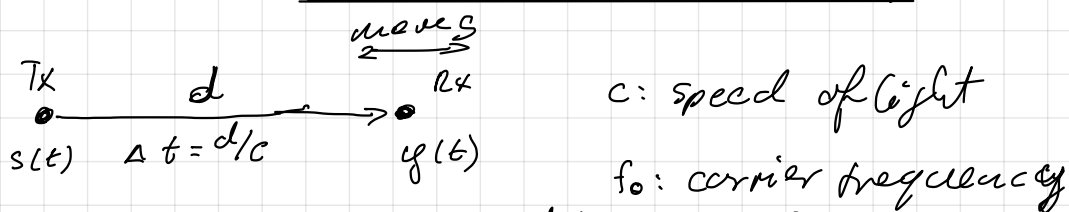


Wireless Receivers

[23.10.2020]

Fading Channels / Diversity



$$\begin{aligned} y(t) &= s\left(t - \frac{d}{c}\right) e^{-j2\pi\left(t - \frac{d}{c}\right)f_0} \cdot e^{-j2\pi f_0 t} \cdot \alpha(d) \\ &= s\left(t - \frac{d}{c}\right) e^{-j2\pi f_0 \frac{d}{c}} \cdot \alpha(d) \quad \text{Wave length: } \lambda = c/f_0 \\ &= s\left(t - \frac{d}{\lambda}\right) e^{-j2\pi \frac{d}{\lambda}} \cdot \alpha(d) \end{aligned}$$

Attenuation and phase depend on d

Example: $\alpha(d) \propto \frac{1}{d^2}$ (free space model) $f_0 = 2.4 \text{ GHz}$

move from $d_0 = 10 \text{ m}$ to $d_1 = 10.1 \text{ m}$ ($\lambda = 12.5 \text{ cm}$)
(+10 cm @ 10 m)

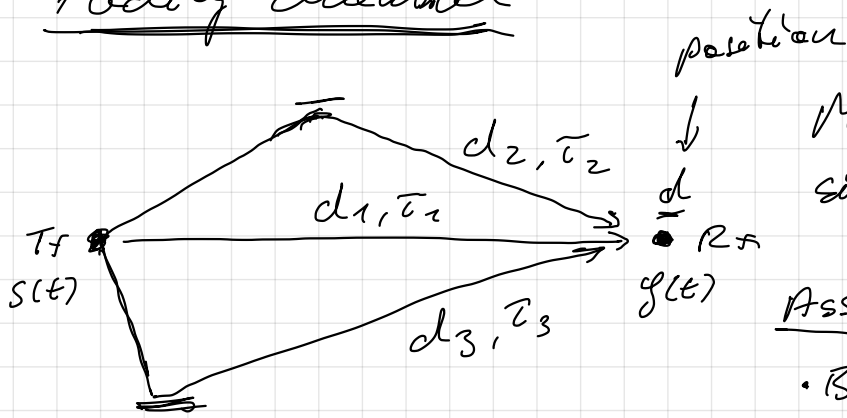
$$\begin{aligned} \frac{\alpha(d_1)}{\alpha(d_0)} &\approx 0.98 \quad (2\% \text{ loss in signal strength}) \\ &\approx 0.09 \text{ dB} \rightarrow \text{no problem} \end{aligned}$$

Phase shift: 288° phase shift

For small changes in the distance we can neglect the impact on signal strength, but not the impact on the phase

If I can track the phase, why does this bother me?

Fading channel



Assume: $d_1 \approx d_2 \approx d_3 \dots$

• But still $|d_i - d_j| \approx \lambda$

• $\frac{|d_i - d_j|}{c} \ll T$

Time of arrival diff. between paths is much smaller than symbol interval

$$y(t) = \frac{1}{N} \sum_{n=1}^N \underbrace{\alpha(d_n)}_{\alpha(\underline{d}) \approx s(t)} \underbrace{s(t - \tau_n)}_{\approx s(t)} e^{-j\phi_n}$$

$$\approx s(t) \alpha(\underline{d}) \frac{1}{N} \sum_{n=1}^N e^{-j\phi_n(\underline{d})}$$

$h(\underline{d})$: channel gain

$$y(t) = h(\underline{d}) s(t)$$

For ~~any~~ given $\underline{d} \rightarrow h(\underline{d})$ deterministic

$$y(t) = h s(t)$$

For any $\underline{d}$: h = Random Variable with some distribution

Implications: channel is a random var

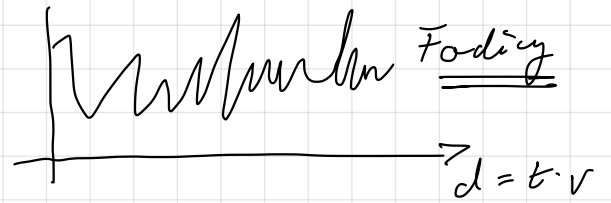
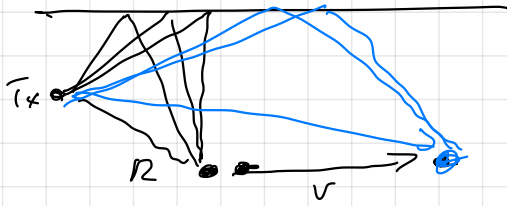
Performance can be evaluated for:

a) a given position/channel $h(\underline{d})$

b) on average across the dist. of that rand. variable h

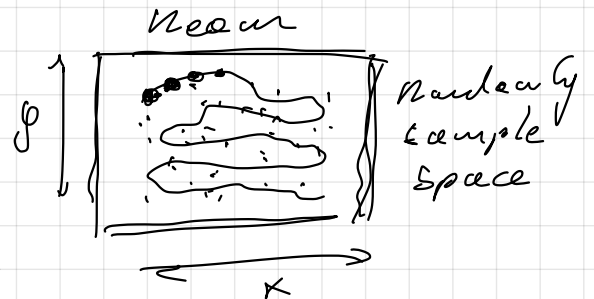
Changing channel

- For a given pos. the channel is fixed
- Moving changes the channel

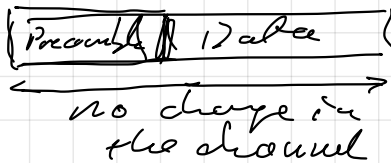
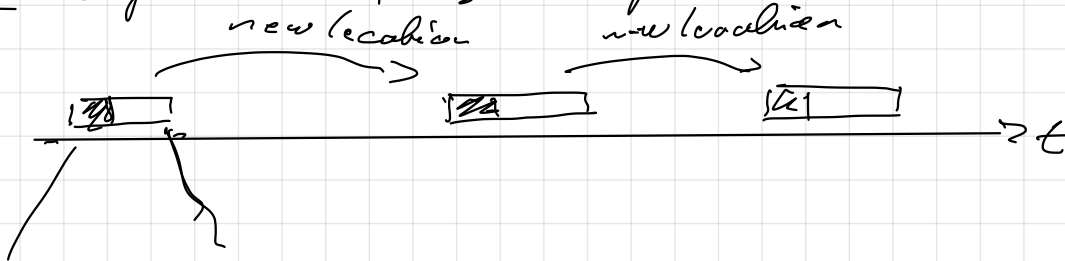


We assume block fading

- Channel remains constant for a long time
- Changes suddenly completely



Tx: is organized in short packets

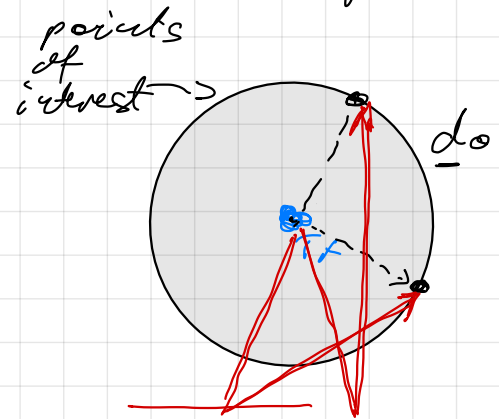


Analyze the performance of the system on average across space

- We need to understand the distribution of h

$$\Rightarrow \underline{d(d)} \equiv \text{const}$$

$$\underline{h} = \frac{1}{N} \underline{d(d_0)} \sum_{n=1}^N e^{j\phi_n}$$



Average SNR:

$$y = hs + w$$

$$P_N: E\{|w|^2\} = \sigma_w^2$$

$$P_S: E\{|hs|^2\} = \underbrace{E\{|s|^2\}}_{P_S=1} \cdot E\{|h|^2\} = 1$$

$$\text{Note: } \frac{1}{N} \int_{d \in d_0} \left| \sum_{n=1}^N e^{j\phi_n(d)} \right|^2 dd = \frac{1}{N} \int \underbrace{\sum_{n=1}^N |e^{j\phi_n(d)}|}_{1}^2 dd = 1$$

Fading does not influence the average SNR

Example: Imagine 2 tributaries (2-paths)

$$1) \phi_1 = +\pi \quad \phi_2 = -\pi$$

$$h = \frac{1}{2} \underline{d(d_1)} \underbrace{(e^{j\pi} + e^{-j\pi})}_0 = 0 \quad \leftarrow \text{no reception}$$

$$2) \phi_1 = +\pi \quad \phi_2 = \pi$$

$$h = \frac{1}{2} \underline{d(d_2)} \underbrace{(e^{j\pi} + e^{j\pi})}_{-2} = -1$$

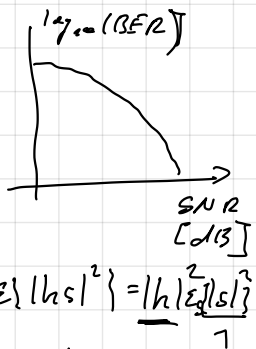
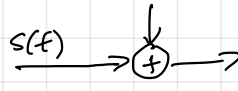
Average BER performance (for a given average SNR)

Reminder: AWGN

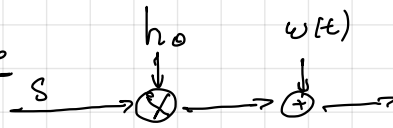
$$w(t) \sim \mathcal{CN}(0, \sigma_w^2), P_s = 1, \overline{\text{SNR}} = \frac{1}{\sigma_w^2}$$

$$P_e = Q(\sqrt{\overline{\text{SNR}}})$$

→ exponential decrease in BER with SNR



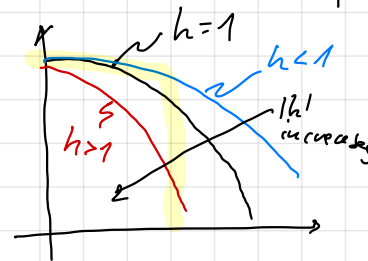
For a given channel h_0



$$y = h_0 s + w$$

$$\overline{\text{SNR}}_{\text{eff}} = \overline{\text{SNR}}(h_0) = |h_0|^2$$

$$P_e(h_0) = Q(\sqrt{|h_0|^2 \overline{\text{SNR}}})$$



Average BER over different h ?

$$E_h \{ P_e(h) \} = E_h \{ Q(\sqrt{|h|^2 \overline{\text{SNR}}}) \} \neq Q(\sqrt{E_h \{ |h|^2 \overline{\text{SNR}} \}})$$

$$\approx 0 \cdot \Pr(|h|^2 \overline{\text{SNR}} = \text{Good})$$

$$+ 0.5 \cdot \Pr(|h|^2 \overline{\text{SNR}} = \text{Bad}) \quad \leftarrow \text{outage event}$$

Distribution of the channel h

$$h = \frac{1}{N} \sum_{n=1}^N e^{j\phi_n} \quad \begin{matrix} \uparrow \\ \text{all} \\ \text{independent} \end{matrix}$$

$N \rightarrow \infty$
CLT

$$h = h_r + h_i j$$

$$h_r \sim \mathcal{N}(0, \frac{1}{2})$$

$$h_i \sim \mathcal{N}(0, \frac{1}{2})$$

$$\sum R_n e^{j\phi_n}$$

$$\sum I_n e^{j\phi_n}$$

$$E\{|h|^2\} = 1 = \underbrace{E\{|h_r|^2\}}_{1/2} + \underbrace{E\{|h_i|^2\}}_{1/2}$$

Polar coordinates $h = |h| e^{j\phi_h}$

• Phase is random between $+\pi$ and $-\pi$
Uniform

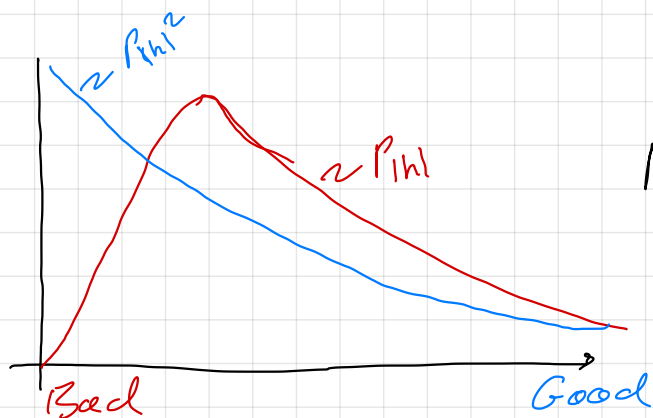
• Magnitude: RAYLEIGH distributed
→ Rayleigh fading
channel

$$P_{\phi_h} = \frac{1}{2\pi}$$

$$P_{|h|} = |h| \cdot 2 \cdot e^{-|h|^2}$$

$$P_{|h|^2} = e^{-|h|^2}$$

Rayleigh distribution



$P_{|h|^2} = e^{-|h|^2}$: χ^2 distribution with 2 degrees of freedom

$$|h|^2 = |h_r|^2 + |h_i|^2$$

$\uparrow$ $\uparrow$ $\uparrow$
 χ^2_2 χ^2_1 χ^2_1

Channel magnitude can vary completely with time probability

Approx of the error rate

$$Q\{\sqrt{|h|^2 \text{SNR}}\} = \begin{cases} 0, & |h|^2 \text{SNR} > \lambda \\ 0.5, & |h|^2 \text{SNR} \leq \lambda \end{cases}$$

- Define the event that $|h|^2 \text{SNR} \leq \lambda$ as outage where we are in a "deep fade"

$$E\{P_e(h)\} \approx \Pr(|h|^2 < \frac{\lambda}{\text{SNR}})$$

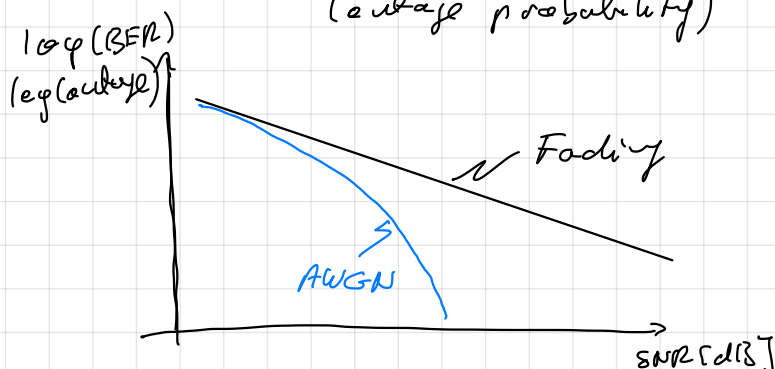
Outage probability is given by fading statistics

$$\Pr(|h|^2 < \frac{\lambda}{\text{SNR}}) = \Pr(|h| < \sqrt{\frac{\lambda}{\text{SNR}}}) = \int_0^{\sqrt{\frac{\lambda}{\text{SNR}}}} |h| \cdot 2 \cdot e^{-|h|^2} d|h|$$

$\sqrt{\frac{\lambda}{\text{SNR}}}$ $\rightarrow 1$ for small $|h|$

$$\approx \int_0^{\sqrt{\frac{\lambda}{\text{SNR}}}} 2|h| d|h| = \frac{1}{|h|^2} \Big|_0^{\sqrt{\frac{\lambda}{\text{SNR}}}} = \frac{\lambda}{\text{SNR}}$$

Observation: BER decays only as $\frac{1}{\text{SNR}}$ (outage probability)



Problem: Significant probability that the channel is in outage